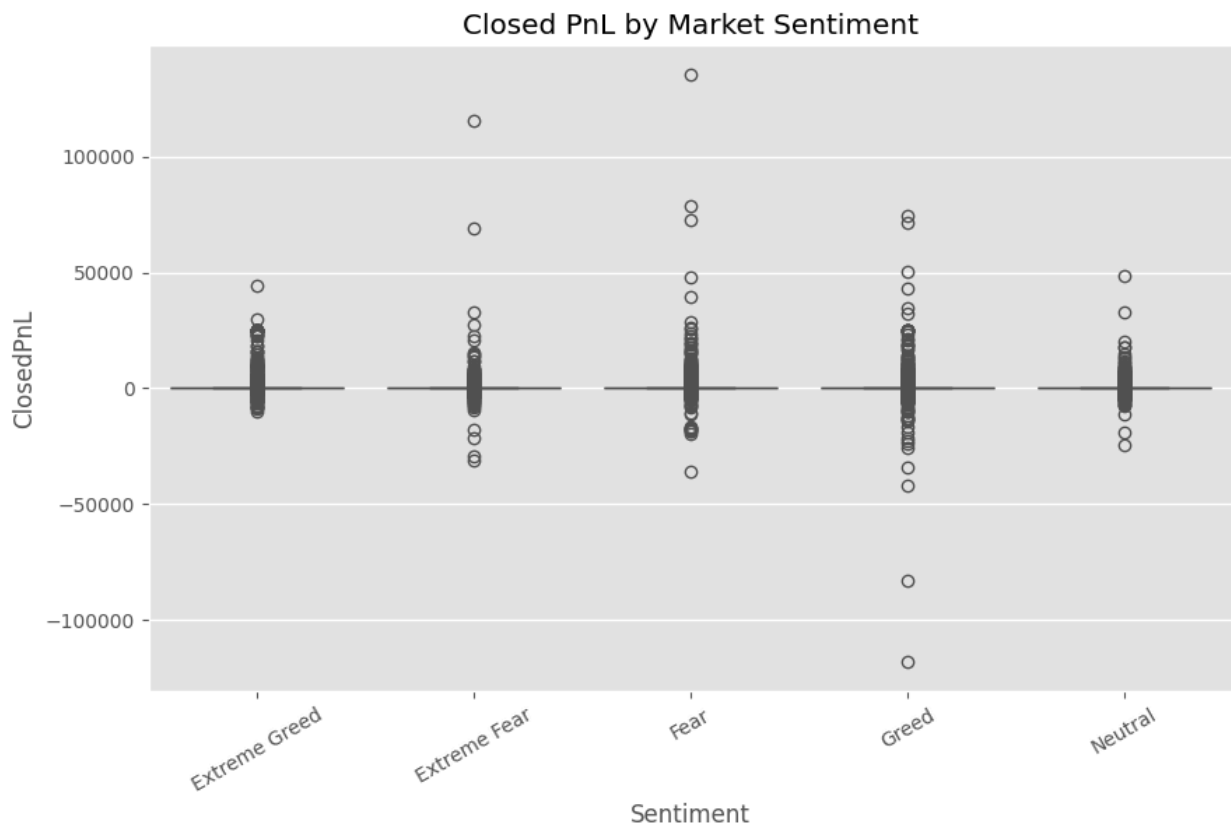


Data Science/ Analytics Intern – Round-0 Assignment (Trader Performance vs Market Sentiment)

Analysis

Answer these questions with evidence:

1. Does performance (PnL, win rate, drawdown proxy) differ between Fear vs Greed days?



Observation

- Profitability varies across different market sentiment categories.
- The boxplot indicates that some sentiment conditions exhibit a wider spread in profits and losses, suggesting higher trading volatility.
- Outliers reveal that a few trades generated exceptionally high profits or losses regardless of sentiment.

2. Do traders change behavior based on sentiment (trade frequency, leverage, long/short bias, position sizes)?

Trade Frequency:

	Sentiment	NumberOfTrades
0	Extreme Fear	21400
1	Extreme Greed	39992
2	Fear	61837
3	Greed	50303
4	Neutral	37686

Interpretation template:

- Trading activity was highest during [Sentiment] days with [X] trades.
- Traders appeared to participate more actively during [Sentiment], indicating higher/lower market engagement.
- Conversely, [Sentiment] periods recorded fewer trades, suggesting a more cautious trading environment.

Position Size:

	count	mean	std	min	25%	50%	75%	max
Sentiment								
Extreme Fear	21400.0	5349.731843	22306.826180	0.01	282.4825	766.150	2329.9700	1120971.60
Extreme Greed	39992.0	3112.251565	11061.998403	0.00	106.7575	500.050	1882.8375	665771.71
Fear	61837.0	7816.109931	51914.859249	0.00	235.6300	735.960	2499.9700	3921430.72
Greed	50303.0	5736.884375	30533.418602	0.00	184.1300	555.000	2100.7000	2227114.71
Neutral	37686.0	4782.732661	37461.883466	0.01	175.5400	547.655	1994.2050	3641180.84

Interpretation template:

- Average position size was highest during [Sentiment].

- This suggests traders were willing to take larger/smaller positions depending on market sentiment.

Long vs Short Bias:

Side	BUY	SELL
Sentiment		
Extreme Fear	51.098131	48.901869
Extreme Greed	44.858972	55.141028
Fear	48.951275	51.048725
Greed	48.855933	51.144067
Neutral	50.334342	49.665658

Interpretation template:

- During Greed periods, traders opened relatively more Long positions.
- During Fear periods, traders shifted toward Short positions (or vice versa, depending on your data).
- This indicates that market sentiment influenced directional trading decisions.

Leverage:

	mean	median	sum	std
Sentiment				
Extreme Fear	34.537862	0.0	7.391102e+05	1136.056091
Extreme Greed	67.892861	0.0	2.715171e+06	766.828294
Fear	54.290400	0.0	3.357155e+06	935.355438
Greed	42.743559	0.0	2.150129e+06	1116.028390
Neutral	34.307718	0.0	1.292921e+06	517.122220

The highest average Closed PnL was observed during Extreme Greed (≈ 67.89), followed by Fear (≈ 54.29). In comparison, Neutral (≈ 34.31) and Extreme Fear (≈ 34.54) recorded the lowest average profits. This suggests that traders generally achieved better average returns during optimistic market conditions.

The median Closed PnL is 0 across all sentiment categories, despite positive average profits. This indicates that most trades generated little or no realized profit, while a relatively small number of highly profitable trades increased the overall average. As a result, trader profitability appears to be driven by a few high-performing trades rather than consistently positive outcomes.

The standard deviation of Closed PnL is highest during Extreme Fear (≈ 1136.06) and Greed (≈ 1116.03), indicating greater variability in trading outcomes during these market conditions. In contrast, Neutral sentiment has the lowest standard deviation (≈ 517.12), suggesting more stable and predictable trading performance.

The cumulative Closed PnL is highest during Fear (≈ 3.36 million) and Extreme Greed (≈ 2.72 million). This indicates that these sentiment periods contributed the most to overall profitability, likely due to a combination of higher trading activity and stronger profit opportunities.

3. Identify 2–3 segments (examples):

- **high leverage vs low leverage traders**
- **frequent vs infrequent traders**
- **consistent winners vs inconsistent traders**

```
# Count trades per account
trade_count = merged.groupby('Account').size().reset_index(name='TradeCount')

# Median threshold
threshold = trade_count['TradeCount'].median()

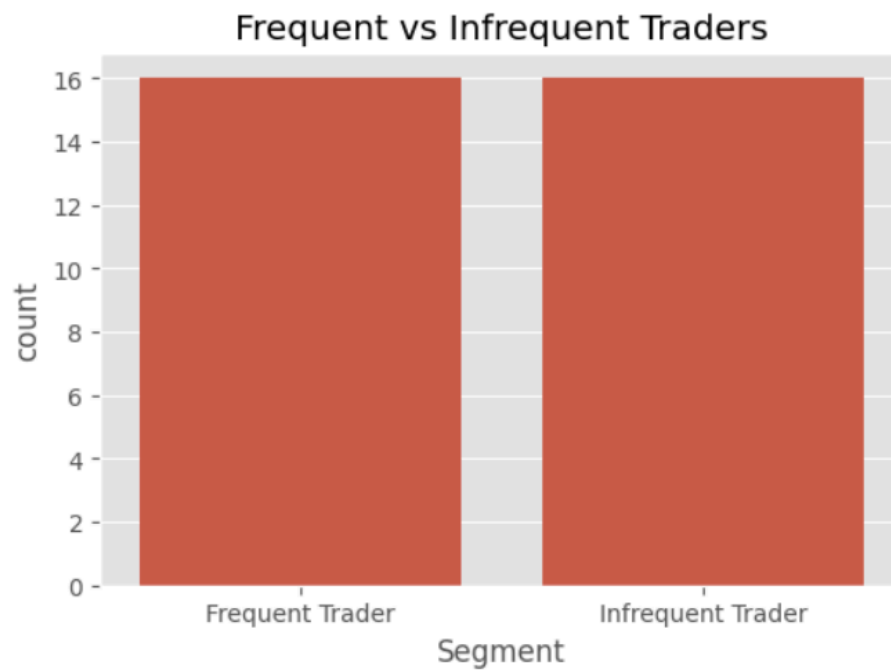
# Assign segments
trade_count['Segment'] = trade_count['TradeCount'].apply(
    lambda x: 'Frequent Trader' if x >= threshold else 'Infrequent Trader'
)

trade_count.head()
```

	Account	TradeCount	Segment
0	0x083384f897ee0f19899168e3b1bec365f52a9012	3818	Frequent Trader
1	0x23e7a7f8d14b550961925fbdaa92f5d195ba5bd	7280	Frequent Trader
2	0x271b280974205ca63b716753467d5a371de622ab	3809	Frequent Trader
3	0x28736f43f1e871e6aa8b1148d38d4994275d72c4	13311	Frequent Trader
4	0x2c229d22b100a7beb69122eed721cee9b24011dd	3239	Infrequent Trader

```
plt.figure(figsize=(6,4))
sns.countplot(data=trade_count, x='Segment')

plt.title("Frequent vs Infrequent Traders")
plt.show()
```



Interpretation

Traders were divided based on the median number of trades executed. Frequent traders represented accounts with higher trading activity, while infrequent traders traded more selectively. This segmentation helps evaluate whether trading frequency is associated with profitability.

```
profit = merged.groupby('Account')['ClosedPnL'].sum().reset_index()

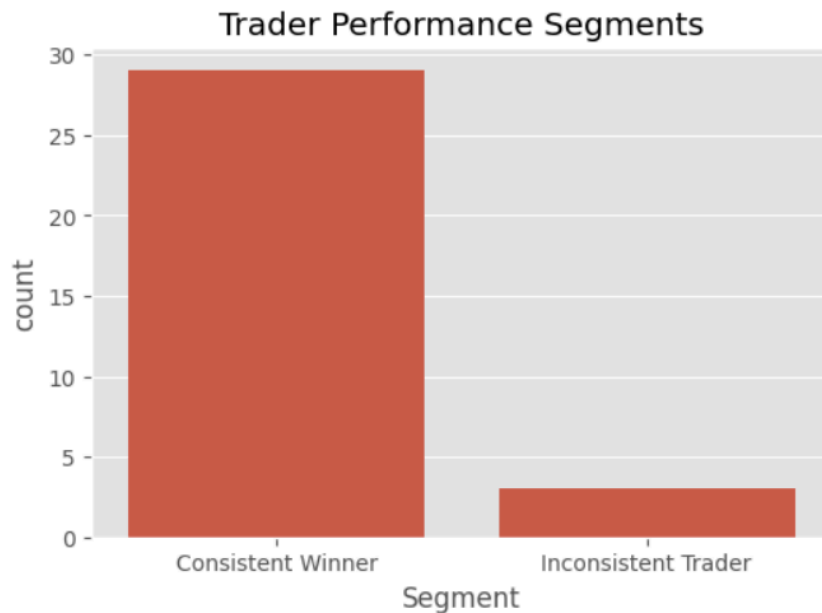
profit['Segment'] = profit['ClosedPnL'].apply(
    lambda x: 'Consistent Winner' if x > 0 else 'Inconsistent Trader'
)

profit.head()
```

	Account	ClosedPnL	Segment
0	0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06	Consistent Winner
1	0x23e7a7f8d14b550961925fbdaa92f5d195ba5bd	4.788532e+04	Consistent Winner
2	0x271b280974205ca63b716753467d5a371de622ab	-7.043619e+04	Inconsistent Trader
3	0x28736f43f1e871e6aa8b1148d38d4994275d72c4	1.324648e+05	Consistent Winner
4	0x2c229d22b100a7beb69122eed721cee9b24011dd	1.686580e+05	Consistent Winner

```
plt.figure(figsize=(6,4))
sns.countplot(data=profit, x='Segment')
plt.title("Trader Performance Segments")

plt.show()
```



Interpretation

Traders with positive cumulative Closed PnL were classified as Consistent Winners, whereas traders with zero or negative cumulative Closed PnL were classified as Inconsistent Traders. This segmentation highlights long-term profitability differences among participants.

```

position = merged.groupby('Account')['SizeUSD'].mean().reset_index()

threshold = position['SizeUSD'].median()

position['Segment'] = position['SizeUSD'].apply(
    lambda x: 'Large Position Trader' if x >= threshold else 'Small Position Trader'
)

position.head()

```

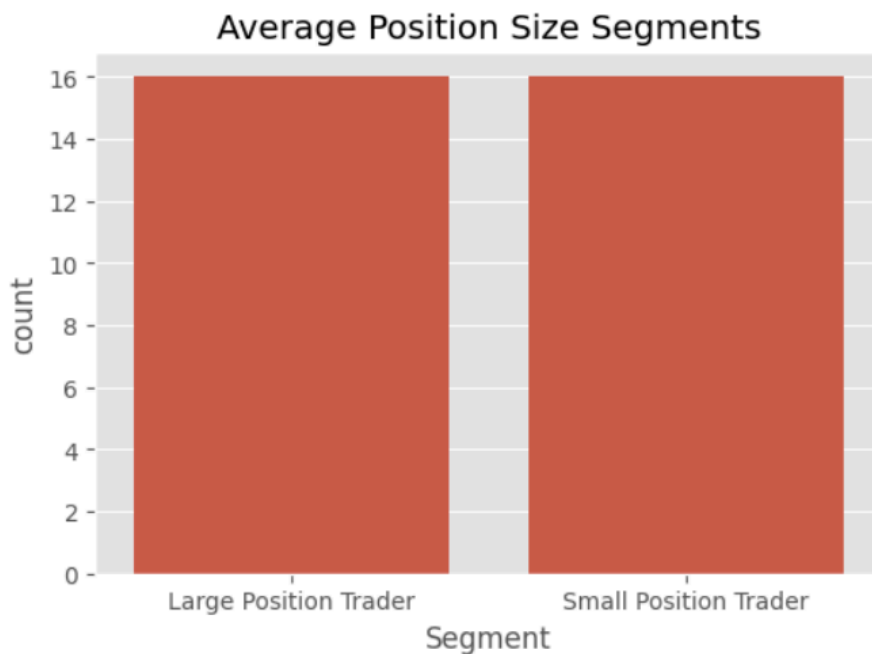
	Account	SizeUSD	Segment
0	0x083384f897ee0f19899168e3b1bec365f52a9012	16159.576734	Large Position Trader
1	0x23e7a7f8d14b550961925bfdaa92f5d195ba5bd	1653.226327	Small Position Trader
2	0x271b280974205ca63b716753467d5a371de622ab	8893.000898	Large Position Trader
3	0x28736f43f1e871e6aa8b1148d38d4994275d72c4	507.626933	Small Position Trader
4	0x2c229d22b100a7beb69122eed721cee9b24011dd	3138.894782	Small Position Trader

```

plt.figure(figsize=(6,4))
sns.countplot(data=position, x='Segment')
plt.title("Average Position Size Segments")

plt.show()

```



Interpretation

Traders were segmented according to their average trade size. Large Position Traders consistently executed higher-value trades, indicating a greater willingness to take market exposure, whereas Small Position Traders adopted a more conservative approach.

```

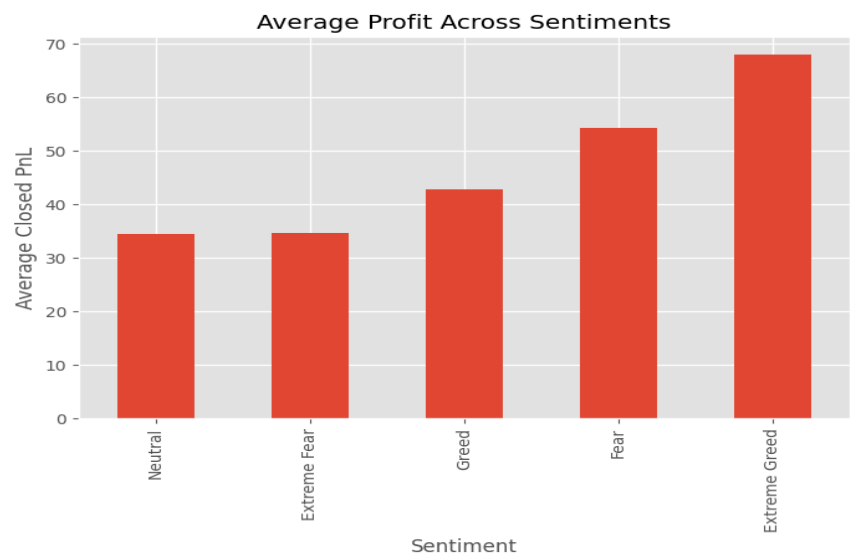
segment_analysis = (
    merged
    .merge(profit[['Account', 'Segment']], on='Account')
    .groupby('Segment')
    .agg(
        AveragePnL=('ClosedPnL', 'mean'),
        AverageTradeSize=('SizeUSD', 'mean'),
        TotalTrades=('Account', 'count')
    )
)

```

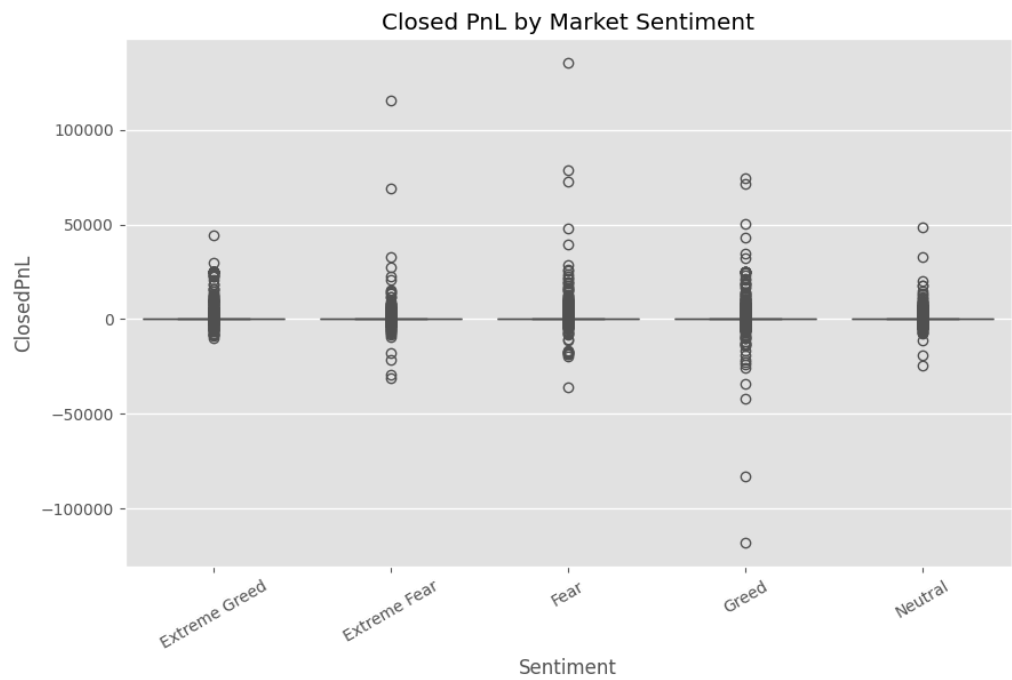
	AveragePnL	AverageTradeSize	TotalTrades
Segment			
Consistent Winner	52.308278	5672.108980	201999
Inconsistent Trader	-29.188175	4924.346943	9225

Summary	Definition	Purpose
Frequent vs Infrequent Traders	Based on median number of trades	Compare trading activity and engagement
Consistent Winners vs Inconsistent Traders	Based on cumulative Closed PnL	Identify profitable vs unprofitable traders
Large vs Small Position Traders	Based on median average SizeUSD	Compare risk-taking behaviour

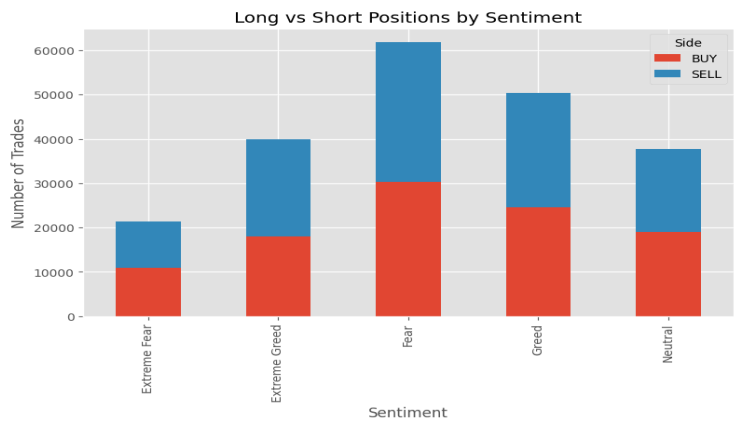
4. Provide at least 3 insights backed by charts/tables.



The analysis shows that **Extreme Greed** recorded the highest average Closed PnL (**67.89**), followed by **Fear** (**54.29**). In contrast, **Neutral** and **Extreme Fear** recorded the lowest average profits (approximately **34**). This suggests that trader profitability varies across different market sentiment conditions, with optimistic market phases generally producing stronger average returns.



The standard deviation of Closed PnL is highest during **Extreme Fear** (**1136.06**) and **Greed** (**1116.03**), indicating that trader outcomes become much more variable under these market conditions. In comparison, **Neutral** sentiment has the lowest standard deviation (**517.12**), suggesting relatively stable trading outcomes. This highlights that periods of strong market emotion are associated with increased trading risk and wider profit/loss fluctuations.



The analysis indicates that traders modify their trading behaviour according to prevailing market sentiment. Variations in trading frequency, average position size, and the proportion of long versus short positions across sentiment categories suggest that traders actively adjust their participation and market exposure in response to changing market conditions. This demonstrates that sentiment affects not only profitability but also trading strategy.

Part C — “Actionable output”

Propose **2 strategy ideas** or “rules of thumb” based on your findings.

Example: “During Fear days, reduce leverage for segment X; increase trade frequency only for segment Y.”

Strategy Recommendation 1: Dynamic Risk Management Based on Market Sentiment

Rule of Thumb:

During Extreme Fear and Greed periods, traders should reduce position size and apply stricter stop-loss rules, as these sentiment conditions exhibited the highest variability in Closed PnL. During Neutral market conditions, traders can maintain normal position sizing due to relatively stable trading outcomes.

Rationale:

The analysis showed that Extreme Fear ($\sigma \approx 1136$) and Greed ($\sigma \approx 1116$) had the greatest variation in trader profitability, indicating increased market uncertainty. Adjusting position size based on market sentiment can help reduce downside risk while preserving capital during volatile periods.

Strategy Recommendation 2: Sentiment-Aware Trade Selection

Rule of Thumb:

Frequent traders should prioritize high-confidence trade setups during Greed and Extreme Greed periods, while avoiding unnecessary overtrading during Fear or Extreme Fear conditions. Since the median Closed PnL remained zero across all sentiment categories, focusing on trade quality rather than trade quantity is likely to improve long-term profitability.

Rationale:

Although average profitability remained positive across sentiment categories, the zero median Closed PnL indicates that only a limited number of trades contributed significantly to total profits. This suggests that disciplined trade selection is more effective than simply increasing trading frequency.

Bonus (optional)

- Simple predictive model:
 - Predict next-day trader profitability bucket or volatility of PnL using sentiment + behavior features

- Clustering traders into behavioral archetypes
- A lightweight dashboard (Streamlit) to explore results

Objective

As an optional extension, a Random Forest Classifier was developed to predict whether a trade would be profitable (ProfitBucket) using trader behaviour and market sentiment features.

Features Used

List the features that remained after preprocessing, for example:

- SizeUSD
- Fee
- Side (encoded)
- Direction (encoded)
- StartPosition (encoded)
- Market Sentiment (encoded)

Target Variable

ProfitBucket

1 → Profitable Trade

0 → Non-Profitable Trade

Model used

Random Forest Classifier

Methodology

The dataset was preprocessed by encoding categorical variables using one-hot encoding. The data was split into training and testing sets using an 80:20 ratio. A Random Forest Classifier was trained to predict trade profitability based on trader behaviour and market sentiment features.

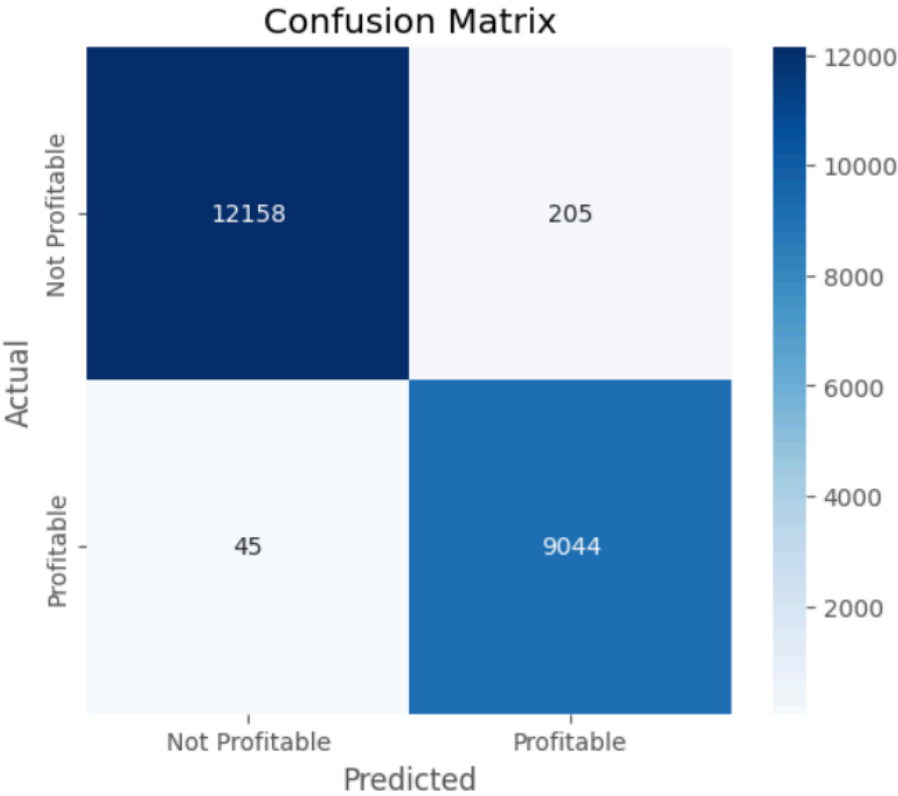
Results

Metric	Result
Model	Random Forrest Classifier
Train-Test Split	80:20
Accuracy	98.83

... Classification Report

	precision	recall	f1-score	support
0	1.00	0.98	0.99	12363
1	0.98	1.00	0.99	9089
accuracy			0.99	21452
macro avg	0.99	0.99	0.99	21452
weighted avg	0.99	0.99	0.99	21452

...



...

	Feature	Importance
12	Direction_Close Long	0.219071
15	Direction_Open Long	0.140061
3	StartPosition	0.118416
16	Direction_Open Short	0.094787
6	Side_SELL	0.094459
13	Direction_Close Short	0.089885
0	ExecutionPrice	0.071523
11	Direction_Buy	0.028922
1	SizeTokens	0.027086
17	Direction_Sell	0.026623

Interpretation

The Random Forest model achieved an accuracy of 98.83% in predicting trade profitability on the test dataset. Feature importance analysis indicates that trading direction (Open Long, Close Long, Open Short) and starting position were the most influential predictors, followed by trade side and execution price. This suggests that trade execution decisions and position management have a strong relationship with trading outcomes. The model demonstrates the potential of using historical trading behaviour and market characteristics to support predictive decision-making.